Unnamed_Unit

Unit 3 - Designing Solutions

PDF Documents

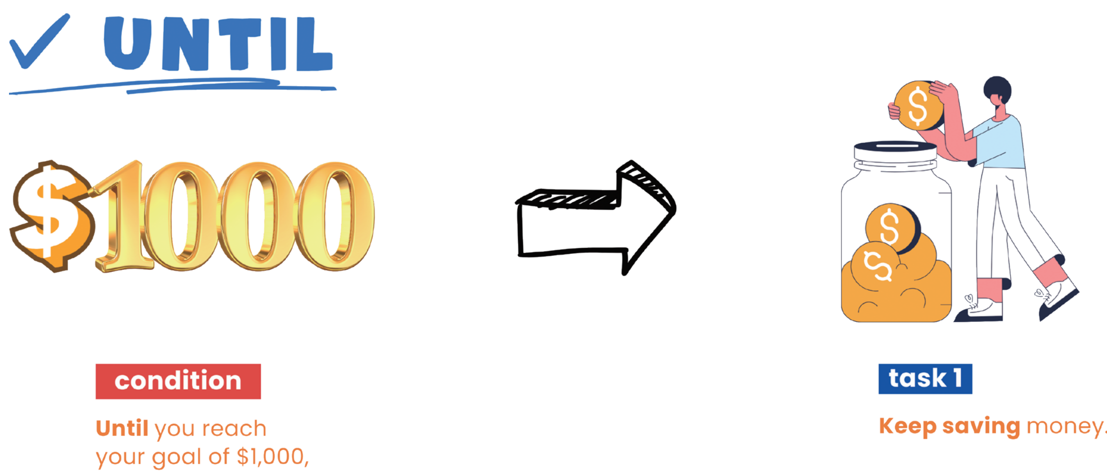
PDF Document 1: 1738150787778-Chapter 8 - Unit 3 - Designing Solutions.pdf

This PDF document is included in the unit materials.

UNIT 3 DESIGNING SOLUTIONS

Until statement

In programming, the "until" statement is like having a goal and doing something over and over until you reach that goal.



For example, if your goal is to save $1,000, you keep adding money to your savings "until" you have $1,000. Just like saving money bit by bit until you hit your target, the "until" statement repeats an action until a certain condition is met.

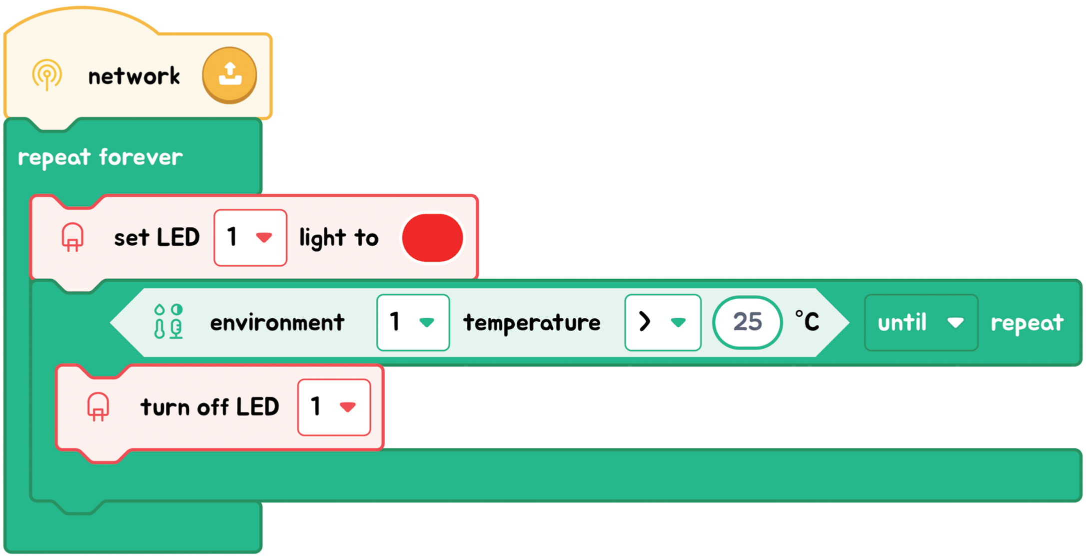
MORE ABOUT "UNTIL"

Think about it: If you plan to study until 9 o'clock, what are the conditions and tasks involved in this plan?

In programming, the condition for this plan is "until 9 o'clock," and the task is "study." The "Until" loop operates based on a condition that specifies when to stop, such as reaching a certain time or event. Without this condition, the code wouldn't know when to conclude the task.

Another Example:

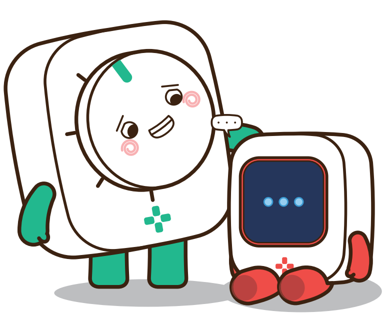
In our example, think of the LED as a light that stays off "until" it gets warm enough, say 25 degrees. So, the LED doesn't turn on while it's cooler.



But once it hits 25 degrees or more, the red light comes on. The "until" statement here waits for the temperature to go up, and when it does, it's like a signal to turn the red light on

CONCLUSION

In brief, the "Until" block executes repeatedly until the condition is met. We'll use this principle to create an automatic door with the MODI modules!



Flowcharts & Algorithms

Let's read the flowchart and create the code.

FLOWCHART